Formulaire d'intégration du champ de déformation

Champ de déformation initial

$\varepsilon_{11} =$	.....
$\varepsilon_{22} =$	.....
$\varepsilon_{33} =$	.....
$\varepsilon_{12} = \varepsilon_{21} =$	.....
$\varepsilon_{23} = \varepsilon_{32} =$	.....
$\varepsilon_{31} = \varepsilon_{13} =$	.....

Équations de compatibilité : $\underline{\underline{\mathrm{rot}}}({}^{\mathrm{t}}\underline{\underline{\mathrm{rot}}}\,\underline{\underline{\varepsilon}}) = \underline{\underline{0}}$

$$\omega_{ij,kl} = \omega_{ij,lk} \quad \Leftrightarrow \quad \varepsilon_{ik,jl} - \varepsilon_{jk,il} = \varepsilon_{il,jk} - \varepsilon_{jl,ik}$$

Tenseur et vecteur de rotation

$$\underline{\underline{\omega}} = \begin{pmatrix} 0 & \omega_{12} & -\omega_{31} \\ -\omega_{12} & 0 & \omega_{23} \\ \omega_{31} & -\omega_{23} & 0 \end{pmatrix} = \begin{pmatrix} -\omega_{23} \\ -\omega_{31} \\ -\omega_{12} \end{pmatrix} \wedge$$

Vecteurs translation $\vec{\lambda}$ et rotation $\vec{\Omega}$ de corps rigide

$$\vec{\lambda} = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix} \quad \text{et} \quad \vec{\Omega} = \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

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1. Calcul de $\omega_{ij,k} = \varepsilon_{ki,j} - \varepsilon_{jk,i}$

$\omega_{12,1} = \varepsilon_{11,2} - \varepsilon_{21,1} =$	.....
$\omega_{12,2} = \varepsilon_{21,2} - \varepsilon_{22,1} =$	.....
$\omega_{12,3} = \varepsilon_{31,2} - \varepsilon_{23,1} =$	.....

$\omega_{23,1} = \varepsilon_{12,3} - \varepsilon_{31,2} =$	.....
$\omega_{23,2} = \varepsilon_{22,3} - \varepsilon_{32,2} =$	.....
$\omega_{23,3} = \varepsilon_{32,3} - \varepsilon_{33,2} =$	.....

$\omega_{31,1} = \varepsilon_{13,1} - \varepsilon_{11,3} =$	.....
$\omega_{31,2} = \varepsilon_{23,1} - \varepsilon_{12,3} =$	.....
$\omega_{31,3} = \varepsilon_{33,1} - \varepsilon_{13,3} =$	.....

2. Calcul de ω_{ij} par intégration de $\omega_{ij,k}$

$\omega_{12} =$	$-r+$ .....
$\omega_{23} =$	$-p+$ .....
$\omega_{31} =$	$-q+$ .....

3. Calcul de $u_{i,j} = \varepsilon_{ij} + \omega_{ij}$

$u_{1,1} = \varepsilon_{11} =$	.....
$u_{1,2} = \varepsilon_{12} + \omega_{12} =$	$-r+$ .....
$u_{1,3} = \varepsilon_{13} - \omega_{31} =$	$+q+$ .....

$u_{2,1} = \varepsilon_{21} - \omega_{12} =$	$+r+$ .....
$u_{2,2} = \varepsilon_{22} =$	.....
$u_{2,3} = \varepsilon_{23} + \omega_{23} =$	$-p+$ .....

$u_{3,1} = \varepsilon_{31} + \omega_{31} =$	$-q+$ .....
$u_{3,2} = \varepsilon_{32} - \omega_{23} =$	$+p+$ .....
$u_{3,3} = \varepsilon_{33} =$	.....

4. Calcul de u_i par intégration de $u_{i,j}$

$u_1 =$	$\lambda_1 - rx_2 + qx_3+$ .....
$u_2 =$	$\lambda_2 - px_3 + rx_1+$ .....
$u_3 =$	$\lambda_2 - qx_1 + px_2+$ .....

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5. Vérification de la compatibilité de la déformation $\widehat{\varepsilon}_{ij}$ (calculée à partir de u_i à l'étape 4) $\stackrel{?}{=} \varepsilon_{ij}$

$\widehat{\varepsilon}_{11} = u_{1,1} =$	.....
$\widehat{\varepsilon}_{22} = u_{2,2} =$	.....
$\widehat{\varepsilon}_{33} = u_{3,3} =$	.....

$\widehat{\varepsilon}_{12} = \frac{1}{2}(u_{1,2} + u_{2,1}) =$	.....
$\widehat{\varepsilon}_{13} = \frac{1}{2}(u_{1,3} + u_{3,1}) =$	.....
$\widehat{\varepsilon}_{23} = \frac{1}{2}(u_{2,3} + u_{3,2}) =$	.....